Meridian Library Web Service - Deployment and Operations Runbook

Meridian County Library Service | Digital Services Group | Document reference DSG/OPS/007 | Version 2.3, revised January 2026 | Owner: D. Iremonger, Infrastructure

1. Scope

How the Meridian Library web service is built, released, rolled back and supported. It assumes the reader has access to the county's deployment jump host and to the service repository. It does not cover the back-office library management system, which is a supplier arrangement with its own procedures.

2. Environments

- **Development** - a developer's own machine, on localhost port 4180. npm install then npm start; no configuration is needed.
- **Staging** - host lib-stage-01, reachable internally only. Identical to production and refreshed from the main branch on demand.
- **Production** - host lib-prod-01, public and behind the county proxy. A single virtual machine; there is no second instance.

Each environment is one small virtual machine in the county data centre running the supported long-term Node.js release. nginx fronts the application, terminates TLS, serves the static directory directly and restricts the /staff/ prefix to county address ranges. The application itself listens on port 4180 and is kept alive by the process supervisor.

3. Build

There is no build step. The service is a server-rendered Express application with no bundler, no transpilation and no generated assets: what is in the repository is what is served. "Building" a release means nothing more than checking out a tag and installing the two runtime dependencies.

The only generated file in the repository is the published address list, which is regenerated from the route table with the routes script. This is a manual step and must be run and committed whenever an address is added or removed.

4. Release procedure

Releases go out on a Tuesday morning outside opening hours, or at any time for a fault affecting the public service. The steps are the same either way.

- Confirm the change is merged to the main branch and that the version in the package manifest has been raised. Any change to a rendered page must raise it.
- Tag the release as **v** and push the tag.
- On staging: fetch, check out the tag, install dependencies, restart under the supervisor. Confirm the health address returns ok.
- Walk the release check on staging - the home page, catalogue search, the county holding, a reservation, the staff catalogue entry page and one reporting page.
- Notify the duty officer and the Fenner Road desk that a release is going out.
- On production: repeat the fetch, checkout, install and restart. The restart is under a second

and is not announced to members.
- Confirm the health address, then repeat the release check against production.
- Record the tag, the time and the person releasing in the change log.

5. Rollback

Rollback is a checkout of the previous tag and a restart, and takes about two minutes. There is no database and therefore no data migration to undo: a release is code and nothing else, and the previous code will behave exactly as it did before.

- Check out the previous tag on the affected host.
- Install dependencies (only necessary if the dependency versions changed, which is rare).
- Restart under the supervisor and confirm the health address.
- Repeat the release check.
- Record the rollback and the reason in the change log, and raise a fault so the cause is looked at rather than forgotten.

The decision to roll back rests with the duty officer and does not need to wait for the Digital Services Manager. If in doubt, roll back.

6. Configuration and secrets

The application takes one setting, the listening port, from the environment and defaults it when absent. It holds no credentials of any kind: there is no database, no third-party service and no outbound integration. TLS certificates are held by nginx and renewed by the county's central process. There is nothing in this service that belongs in a secret store.

7. Monitoring and alerting

The county monitoring polls the health address every minute. Three consecutive failures raise an alert to the duty officer's pager. nginx access and error logs are kept for thirty days; the application's own output goes to the supervisor journal and is rotated weekly. There is no application metrics collection and no error tracking service, so a fault that does not stop the process will not be noticed by monitoring - it will be reported by a member of staff at a branch.

8. On-call

The service is in the county's shared out-of-hours rota. The duty officer covers all Digital Services applications and is not expected to know this one in detail; the expectation out of hours is restart, then roll back, then escalate. Escalation is to the Infrastructure lead and then to the Digital Services Manager.

Because the library is closed outside opening hours, an out-of-hours outage is not treated as urgent unless it falls within the hour before opening. In practice the service is restarted in the morning and the cause looked at during the working day.

9. Routine tasks

- **Node.js version** - moved to the next long-term release each spring, on staging first, with a full walk of the release check.
- **Dependency updates** - the two dependencies are reviewed twice a year and updated only for a security advisory.

- **Events calendar** - carries a fixed month and is edited by hand each September. This is a release like any other.
- **Log review** - the nginx error log is skimmed monthly for repeated not-found addresses, which usually indicate a printed leaflet pointing at an address that no longer exists.

10. Known operational risks

- One production host with no second instance. A host failure is an outage until the host is recovered.
- The staff console is protected only by the proxy restriction. A misapplied nginx configuration exposes it, and nothing in the application would prevent use.
- The reporting pages are publicly reachable by design of the proxy rules as they stand.
- There is no automated test suite, so the release check is walked by hand and its coverage depends on who walks it.